Learning Objective 6.3: I can describe how HD Radio works and demodulate an HD Radio signal.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). Use the hybrid numbers from the lesson: subcarrier spacing $\Delta f = 1{,}488{,}375/4096$ Hz, sidebands from subcarrier 356 to 546 on each side, ten partitions of 19 subcarriers per sideband, and a guard of $7/128$ of the useful symbol time.

1. **Finding the sidebands.** A hybrid station broadcasts at 98.5 MHz.

   (a) Find the frequency of subcarriers 356, 450, and 546, in kHz from the channel center, to 0.1 kHz.

   (b) Between which two radio frequencies, in MHz, should the upper sideband appear on a spectrum display? The lower sideband?

   upper

   lower

   (c) Counting the partition that starts at subcarrier 356 as the first, which partition holds subcarrier 450? The reference subcarriers are 356, 375, 394, and every 19th after that, up to 546. Is subcarrier 451 a reference subcarrier?

2. **Choosing an FFT size.** You record the band with an RTL-SDR at 2.4 MS/s and look at it with a QT GUI Frequency Sink.

   (a) Find the bin width and the number of bins one 69,041 Hz sideband spans, for an FFT of 2048 points and of 16384 points.

   2048

   16384

   (b) Which FFT size can locate a sideband's edge to within one 363.4 Hz subcarrier? Which one would you use to find the sidebands quickly on a live band, and why? Why does neither show the subcarriers as separate peaks?

3. **OFDM timing and data rate.**

(a) Find the useful symbol time T_u, the guard T_g, the whole symbol time, and the symbol rate.

T_u ____

T_g ____

R_s ____

(b) Each data subcarrier carries one QPSK symbol per OFDM symbol. Find the raw bit rate of the 360 data subcarriers in both sidebands.

____

(c) The hybrid mode delivers about 98 kbit/s of program data. What fraction of the raw rate is that, and what do you think the rest carries?

(d) A reflection from a mountain travels 30 km farther than the direct path. Find its delay. Does it fit inside the guard? How many symbols would it span if the sideband were one carrier with a 14.5 μs symbol?

τ ____

symbols ____

4. **Sideband power.** An FM station's analog signal is 6 kW effective radiated power.

   (a) At -20 dBc, find the total digital power in watts, the power of each sideband in watts and in dBc, and the power of each of the 382 subcarriers in dBc.

   total

   sideband

   subcarrier

   (b) The station raises its digital power to -10 dBc total. Repeat (a).

   (c) The L14 meters read -26.5 dBc for the upper sideband and -23.1 dBc for the lower on another station. Find the total digital power in dBc, and say what the difference between the two tells you.

   (d) On a weak station the shelves stand only 10 dB above the noise floor. Does the meter read the sideband high or low, and by about how much? Why does the frequency sink show the shelves about 25 dB under the host's peak when each sideband is -23 dBc?